Synthesising Touch: Haptic-rendered Practice Carillon

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Abstract

This paper describes the design and construction of a prototype haptic carillon baton, and mathematical modelling of the carillon mechanism.
Other research which haptically renders the grand piano mechanism inspires analysis of the kinematic constraints of the carillon mechanism. Analysis is used to construct a physical model using Simulink.
This is then implemented numerically in a Java application. A microcontroller is programmed to interface the prototype's motor and force sensor with a desktop Java application, allowing realtime simulation of the computational model in conjunction with the prototype.
A strategy for containing all physical model computations on an AVR Microcontroller is outlined; this is designed to allow stand-alone operation of the carillon, removing the need for any other external computing hardware.

Introduction

Haptically rendered instruments are designed to remove a major flaw in otherwise useful 'practise' instruments; namely, the absence of an authentic sense of touch, or 'feel', which accompanies any instrumental interaction.

'Feel' can be simply defined as force felt by a player at the point of contact with an instrument; a brass player feels the vibration of his lips, a violinist experiences vibration and resistance at the point where the bow is held, a pianist feels different levels of resistance at the key, etc.

It is possible to predict and recreate (at least, theoretically) forces felt by a player at the point of contact with an instrument by analysing its mechanical properties. This requires an understanding of how the mechanical components of a particular instrument interact, or a *kinematic* analysis, prior to considering the effects of user input.

This type of analysis is already being used to develop physically modelled synthesis algorithms; the interactions between physical components that contribute to sound production are expressed as equations that model an instrument's response to an excitation from a player. A physical model for the synthesis of a violin, for example, will consider the interaction of the bow against the strings, the width of the bow, the damping and resonance of the string, the transfer of energy through the bridge, and the resonance of the soundboard.

A kinematic analysis with a view to haptically rendering an instrument, however, looks at the interactions between physical components that contribute to the force felt by the user. The kinematic constraints of a brass instrument include, for example, the width and depth of the mouthpiece, the length of the tube, and the type of metal used.

In haptically rendered instruments such as the vBow (Nichols 2002) and the Touchback Keyboard (Gillespie 1994, 1996), physical models compute the behaviour of their respective traditional instrument's mechanical systems under different excitations, or gestures, performed by a player. Using sensors (force, position sensors etc.) to monitor a player's gestures, the computational models are able to determine what a player might expect to feel in response, and then actuate this response using a motor.

A thorough taxonomy of new instruments which are either originally conceived, or emulate traditional instruments can be found in the author's previous work (Havryliv et al. 2006).

Mechanics of Grand Piano & Carillon

It is tempting to draw parallels between the conception of a haptic-rendered piano and a haptic-rendered carillon.

The traditional instruments they emulate share a one-dimensional input mechanism (piano: key; carillon: baton) arranged across the instrument in a similar way – i.e. white notes in a bottom row, black notes in the top. Unlike a pianist, however, a carillonneur usually strokes the baton from above with a closed, vertical fist.

In both cases, this one-dimensional input mechanism offers limited scope for controlling timbre: a player may only effect timbre or intensity by controlling the velocity of a striking mechanism (piano: hammer; carillon: clapper) against a vibrating, sound-producing surface (string; bell).

Impact velocity is determined by a player's *gesture*, which is best described as the displacement of the input mechanism over time. This term, gesture, wholly encapsulates the mechanical relationship between a player and an instrument; and it is during the execution of a gesture that a player relies on haptic feedback from an instrument.

It is also in the execution of gesture that differences between the mechanics of a carillon and those of a piano are revealed, and the extent to which pianists and carillonneurs rely on different types of haptic feedback becomes crucial.

Piano

As observed by Gillespie (Gillespie 1996) and others (Oboe et al. 2002), the grand piano mechanism

(Fig. 1) provides a considerable haptic challenge, and as such mechanical and haptic engineers have studied it extensively.

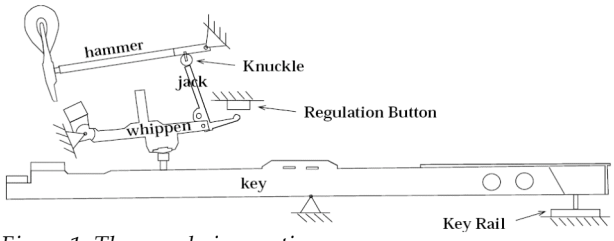


Figure 1. The grand piano action.

The mechanical behaviour of the piano mechanism is characterised by three, discrete phases of different kinematic motions when the key is being pressed downward:

> i) Acceleration – the key is pressed downward but the jack has not yet risen to contact the regulation button;
> ii) Let-off – starts when there is contact between the jack and regulation button, ends when there is no longer compression between the jack and hammer at the knuckle; and
> iii) Free-flight – the hammer is in free-flight towards the string, and the key, whippen and jack remain in motion until the key returns to the key rail.

A primary haptic challenge in this mechanism is rendering the trigger-like feel of the let-off phase. This makes the piano mechanism a system of discrete phases, and the transition between phases of the utmost importance. It is especially crucial for the pianist to have a haptic sense of this let-off transition, as they have no control over the movement of the hammer after this point.

> Our particular interest … lies in the fact that the key and the pianist's finger are completely decoupled from the hammer during its brief period of interaction with the string. The hammer flies free of the jack which initially propels it some 2.5 milliseconds before striking the string. The pianist has no means at his disposal for controlling the tone or the evolution of tone after the hammer has left the jack, except through the damper. Thus all parameters of the tone must be set up by the pianist before tone onset. (Gillespie 1996:9)

Figure 2 displays the motion of a hammer during a gesture which did not impart enough velocity for the hammer to hit the string.

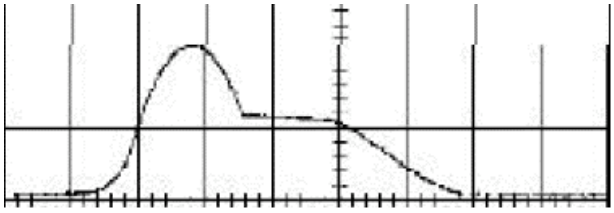


Figure 2. The horizontal line 1/3 of the way up this graph indicates the let-off threshold, i.e. the point at which the hammer flies towards the string; the parabolic trajectory after the let-off demonstrates that the hammer did not impact with the string, but rose and fell according to the laws of simple motion. (Image from Oboe et al. 2002:5)

A corollary to the importance of rendering these transition phases is the relative unimportance of comprehensively rendering the forces felt by a player during non-impact phases. This is evidenced by the common use of position rather than force sensors in haptic piano designs to determine a player's input; the forces felt by a player are closely aligned with the key's position – the mass of the piano mechanism contributes negligibly to the sensation of inertia mid-flight.

This is entirely different to the behaviour of the carillon, in which the combined mass of the mechanism is most certainly a factor that determines the motion of a baton and the force feedback felt by a player.

Carillon

The carillon mechanism (Fig. 3), whilst being a mechanically complex construction, is a simpler kinematic arrangement than that of the piano.

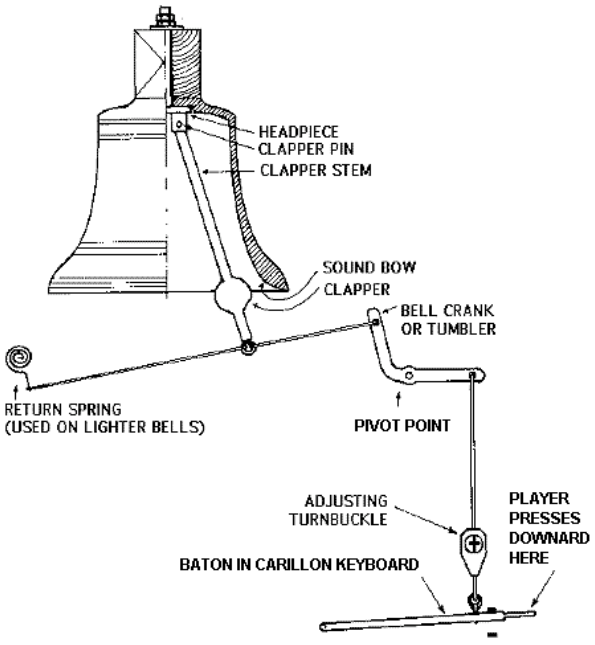


Figure 3. The carillon mechanism. This diagram does not show the complex mechanical arrangements required in a real carillon, but rather a kinematic view of the mechanical interactions which constitute the forces felt by a player.

The above figure identifies the essential elements of the carillon action. A player presses down at the end of the baton, which pulls on one end of a crank, the other end being connected to the clapper. As the crank rotates on a pivot point, the clapper is pulled towards the bell. (Note that the clapper is pulled upwards at an angle, circa 30-45 degrees.) An appropriate kinematic model for this mechanism is the frictionless pulley on an inclined plane of classical Newtonian physics. This is shown in Figure 4 where θ is the angle at which the clapper is pulled up against the inner bell wall, mg sin θ is the negative force applied to the clapper by gravity, Fp is the positive force applied by a player to the clapper and Fnet (i.e. Fp – (mg sin θ)) is the total force acting on the clapper. If Fnet is positive (i.e. the player exerts more force than gravity acting with the clapper's mass does), the clapper will be drawn up against the bell, and if Fnet is negative, the clapper will move away from the bell.

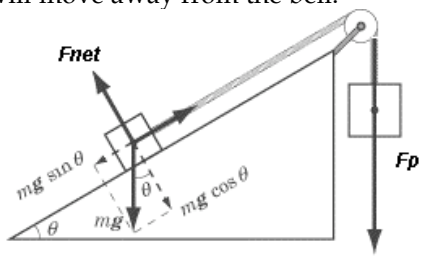


Figure 4. A frictionless inclined plane with pulley, the a simple and expedient model for analysing the forces at work in a carillon mechanism.

Here, it is important to note that we are not primarily concerned with the force felt by a player at the point of contact with the baton, but rather the net force acting on the clapper; this net force is used to determine the displacement of the clapper, and therefore the baton. This simplifies calculations when determining the force applied to the clapper upon collision with the bell.

Physics of the Carillon Mechanism

Continuing the analogy with a frictionless pulley, the small box on the incline represents the clapper, the circular pulley at the top right of the incline is the bell crank, and the larger box hanging down from the pulley is the baton. Note that in the case of the carillon mechanism, a stopper prevents the baton from moving continually upwards and the bell wall prevents the clapper from doing the same (imagine the pulley as an un-passable barrier).

The angle θ is the angle at which the clapper is pulled up against the wall, and is used to determine exactly how much gravitational force is applied to the clapper. In most carillons, this angle changes according to the position of the clapper; the effect of this is addressed further on.

In addition to these basic forces, the carillon mechanism builds in a number of sophisticated mechanical advantages for the player, starting with the baton which allows the player a torque-given advantage over the mass of the clapper of about $1/3^{rd}$; this is covered later in this paper.

Forces at work in the Carillon

There are four main forces in the carillon mechanism: the force of the clapper, the force applied by a player (both represented in Figure 4), the force applied at the impact between the clapper and the bell, and the force applied by the stopper to the baton as it returns to its resting position. These are broken into negative and positive forces:
Negative – clapper force and bell impact force;
Positive – player force, and baton stopper force.

Here, negative forces pull the clapper downwards and pull the baton upwards, whilst positive forces pull the clapper upwards and push the baton downwards. It is useful to note that the respective positions of the clapper and the baton are inversely coupled. i.e. when the clapper has covered $2/3^{rds}$ of the distance upwards towards the bell, the baton is $2/3^{rds}$ of the distance to its bottom stopper.

Clapper and apparatus force: F_C

A constant force is applied to the clapper, this force is the product of its mass, shape and the angle at which it is pulled towards the bell. The angle of the clapper effects the extent to which gravity pulls it downward. The simple equation:

$$F_c = clapperMass * (gravity * sin\theta) \quad (1)$$

gives the appropriate force value. (The right hand side of this equation is labelled in Figure 4 as mg sin θ.)

Additional to the clapper, there are mechanical features which add to the weight felt by the player pressing the baton. These are primarily the masses associated with the transmission mechanism which converts the baton's motion to force applied to the clapper. However they are not analysed at this point as they differ dramatically between different carillons, and even different keys on the same carillon.

Further, the computational model does not need to distinguish between this added mass and the mass of the clapper. Because the angle at which the clapper moves towards the bell changes with position, the capacity to handle non-linearity in the force applied by the clapper has been built into the model.

The testing procedure we have developed considers change in both the clapper's angle and other transient forces as a kind of 'black box' – it is enough to understand their effect on the motion of the baton without a detailed understanding of their construction.

Impact between clapper and bell wall: F_I

Upon impact with the bell, a momentary, yet extremely large force is applied to the clapper; this force propels the clapper downwards, away from the bell. A detailed study of the dynamics of bell clapper impact at the National Carillon, Canberra (Fletcher et al. 2002) provides an excellent reference for modelling this impact, as well as providing useful data relating to bell and clapper masses.

The amount of time the clapper is in contact with the bell during impact determines the force of the collision, as well as the character of the resulting sound. Goldsmith (1960) extends Hertzian impact theory and develops an equation for determining the contact time during an impact between a sphere (clapper) and a very massive plate (bell wall).

The contact time (τ_H) is a function of the clapper's mass (m), the clapper's radius (R), the clapper's elasticity, the velocity of the clapper at impact (V):

$$\tau_H \approx 4.5\left[\frac{(\delta_1+\delta_2)^2 m^2}{RV}\right]^{1/5} \quad (2)$$

where

$$\delta_1 = \frac{1-\mu_1^2}{\pi E_1}; \quad \delta_2 = \frac{1-\mu_2^2}{\pi E_2} \quad (3)$$

and μ_1, μ_2 are the Poisson's ratios (compression on impact) and E_1, E_2 the Young's moduli (elasticity) of the clapper and bell, respectively.[1]

After calculating the contact time, we are able to use this rather extravagant equation to determine the force curve through the impact time:

$$F \approx F_{\max} \sin(\pi t/\tau_H) \quad (4)$$

where

$$F_{\max} \approx 0.44\frac{m^{3/5}R^{1/2}V^{6/5}}{(\delta_1+\delta_2)} \quad (5)$$

However, experimental results demonstrate that a far simpler impulse equation is just as effective in determining the force exerted by the impact:

$$\boldsymbol{F_I = (m(v_f - v_i)) / \tau_H} \quad (6)$$

where v_f is the velocity of the clapper after impact, and v_i is the velocity of the clapper prior to impact.

[1] In the National Carillon, the Poisson ratios are 0.37 & 0.29 for the bell and clapper respectively (the bell being bronze and the clapper, iron). The Young's moduli are 124 & 196.

The difference between the two equations results in a negligible change to the haptic response, and can be classed as unnecessarily complex, especially when compared to the gains in computational efficiency when using the simpler equation.

Debounce; baton stopper force: F_B

Like the impact force between the clapper and the bell, the force resulting from impact between the baton returning to its détente position and the baton stopper is similarly momentary. However, unlike the clapper bell impact force, no comprehensive study has accurately mapped the behaviour between the baton and the baton stopper. The point of contact between the baton and the baton stopper is a variably thick piece of felt, which has made such a collision difficult to model.

Further, the tension in the link between the key and the clapper mechanics may not be calibrated properly, and there will still be residual slackness in the link after the baton has hit the stopper, causing back-and-forth jitter before the baton bounces off. (Clavier calibration is relatively straightforward, and is performed using the adjusting turnbuckle, shown in Figure 3. These may slip even during a performance, and carillonneurs often perform minor adjustments while playing.)

Experiential and impartial observations, however, indicate behaviour very similar to that of a bouncing ball, and this has been implemented in our model – with a parameter to alter the degree of compression present in the felt piece, as one can vary the elasticity of a bouncing ball. Figures 5a and 5b demonstrate the effects of changing this parameter.

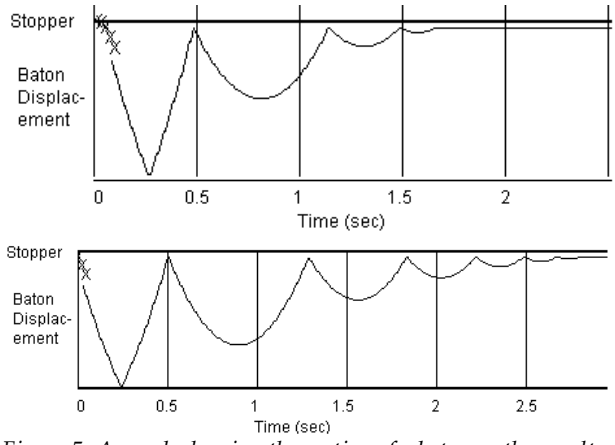


Figure 5: A graph showing the motion of a baton as the result of one gesture. The 'X' points mark the period during which the player is applying force. a) Assumes a thick, dull felt at the baton stopper, absorbing energy and not propelling the baton downwards with as much force as the lighter felt assumed in b). The force applied by a player is the same for both graphs.

Player applied force: F_P

A player may exert a variable force at any time through the performance of a gesture, and this force excites the rest of the system.

Straightforward, linear gestures

If the system is at rest, i.e. the baton is not in motion, a straight forward gesture that the application of a reasonably constant force for the duration of the baton's trajectory will result in a motion like that shown in Figure 6. Figures 6 and 7 show the movement of two clappers, 5kg and 9kg respectively, and are the inverse of baton motion shown elsewhere in this paper.

The following measurements are observed after executing our designed physical model in Matlab, further implementation of this physical model is discussed in a later section.

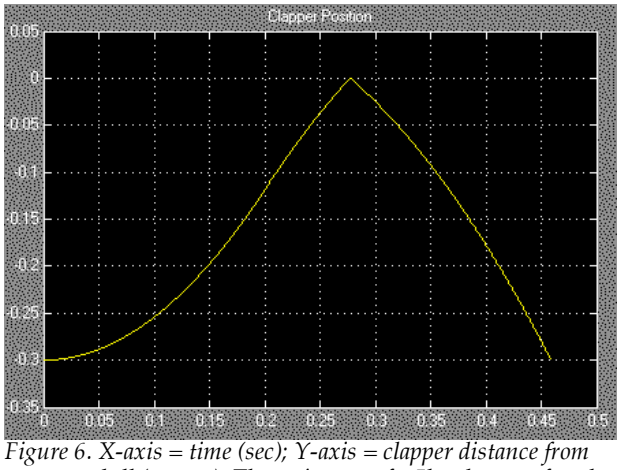


Figure 6. X-axis = time (sec); Y-axis = clapper distance from bell (metres). The trajectory of a 5kg clapper after the application of 80 N of force to the baton for approximately half the baton's full travel distance (until -0.15 on the y-axis).

Depending on the weight of the clapper, applying the same force will result in a different clapper motion. Figure 6 shows a 5kg clapper's motion after a player-exerted force of 80N; Figure 7 shows the motion for a 9kg clapper when the same force is applied for the same period:

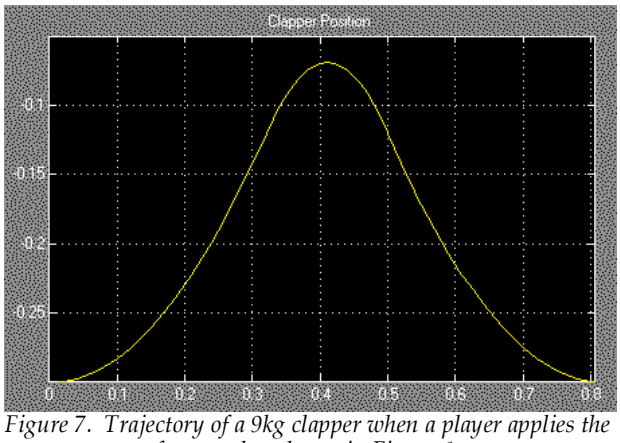


Figure 7. Trajectory of a 9kg clapper when a player applies the same force as that shown in Figure 6.

It is clear from Figures 6 and 7 that differently weighted clappers require different levels of force from a player. In the case of the above example, sufficient force to strike the bell gently and return to the détente position within 0.45 secs in the case of a 5kg clapper was only just enough to carry the 9kg clapper over 2/3^{rds} the distance to the bell, and take 0.8 secs to fully return.

In this execution of simple gestures, the haptic response of the carillon resembles the piano, and indeed, a haptic solution could present itself in the form of a simple resistance mechanism, even springs! It is when more complex gestures, more complex and continuous excitations are employed by a player that the carillon exhibits force-feedback that requires the response capabilities allowed by a realtime physical model.

Complex, continuous gestures

Normal performance practise for carillonneurs involves developing a feel for the mechanical response of the instrument as they play it. It is not only important to develop an attenuation for the initial resistance caused by the mass of a clapper and its associated mechanism. Techniques like tremolo require a performer to interact with continuing forces in the carillon, such as the momentum a clapper builds up as it returns from an impact with the bell. A computational model based on a kinematic analysis makes this reasonably straightforward: Consider Figure 8, in which repeated gestures of the same magnitude and duration are performed by a player and note the difference in response from the system:

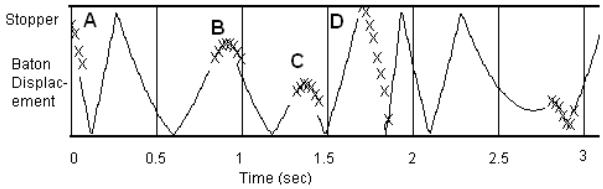


Figure 8. A graph showing the position of a baton through time as a player executes a series of gestures. The 'X' marks show when force is applied by the player to a baton; each gesture is exactly the same application of force over the same period of time.

Figure 8 shows show a series of complex continuous gestures recorded over 3 seconds. Gestures are denoted by 'X' marks. The first gesture is a straightforward attack of considerable force that propels the baton from the stopper with enough impact force to hit the bell again. The gesture at B catches the baton as it rises after the clapper's collision with the bell. This impact poses a counter-force that propels the baton upwards even as the player applies force to it. The same happens with the gesture at C.

The gesture at D illustrates the role of momentum as the baton moves downward from its impact with the stopper, the player adds to the downward force. Note the visual concentration of 'X' marks, which are less dense as the baton travels further during a period of time. The first and fourth gesture occur at the same point in the baton's trajectory. However the fourth gesture has the advantage of force imparted by the collision between the stopper and the baton. Consequently, it propels the baton significantly further and quicker than the

first gesture, which was applied when the baton was stationary.

Implementation of a Carillon model

Simulink, Matlab's GUI solver for its ODE (Ordinary Differential Equation) engine, was used to prototype the physical model derived from the kinematic analysis established in previous sections. Whilst useful in testing the model's performance there are foreseeable budgetary constraints in controlling as many as 53 motor/sensor control systems – as required for the National Carillon. Simulink creates effective microcontroller code from GUI models, however particular supported microcontrollers are themselves costly.

The authors' proclivity towards low-level implementations also played a part in deciding to use a low-cost AVR[1] microcontroller to control the servomotor attached to the prototype (Schiemer et al. 2004). The cost advantages of realising this using open source software also played a part in deciding to develop the realtime physical model engine in Java.

Java and AVR microcontrollers

A desktop Java application was developed which solved the physical model in realtime according to external excitations, i.e. player applied force, test software applied force etc. Rather than using standard ODE solver software already available to Java, it was with a view to implementing the haptic engine for each baton on a dedicated 8-bit microcontroller that the Java engine was written to solve equations numerically, at steps of 1/1000th of a second. This is also the non-realtime rate at which Simulink solved the original implementation.

Furthermore, it seemed sensible to aim towards a microcontroller solution given that a microcontroller will in any case be required to read the force-sensor data from the prototype, as well as controlling the motor.

Java engine

The speed at which realtime physical modelling can be performed is a frustrating constraint; as a solution to this, a Java engine was developed which reduced the physical model to a small number of discrete elements and processes. Shown in Figure 9, separate forces are summed together in order to determine the net force being applied to the clapper. Note the positive and negative forces respectively. This net force is integrated once to determine velocity, then integrated again to determine position. This position data is then sent to the microcontroller which is controlling a motor, the hardware arrangement is discussed in the next section.

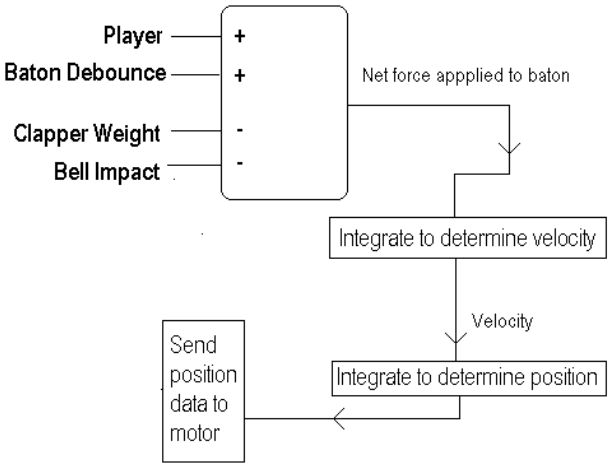


Figure 9. Simplified view of evaluations made by Java engine.

Integration is performed using the standard equations which convert static (non-changing) forces and motions using averaging (Eq. 5); however the very small step size at which these are performed allows *constantly changing* forces and motions to be integrated correctly:

$$\Delta p = v_i t + \tfrac{1}{2} a t^2 \qquad (7)$$

where Δp is the change in position from the last time this equation was solved. The only data this method is required to store between realtime solutions is the previous velocity (v_i), the length of time since it was last solved (t) and the current acceleration (a), which is easily determined from the net force.

Whilst the realtime engine solves every 100 uSecs, it is only polled by the microcontroller every 1-2.5 mSecs; this enables it to solve in quick bursts before waiting a small period to allow time to pass and the microcontroller to request an updated motor position.

AVR and hardware

The AVR ATMega16 microcontroller is used to interface hardware components to the Java desktop physical model engine. Hardware components include a GWServo S04 BBM[2] servomotor and an Interlink[3] force sensor. A servo motor was chosen for its low power consumption, high torque, and because it accepts position, rather than velocity, commands which removes the need for a separate position sensor in the prototype – given that the microcontroller can be reasonably certain that the motor is in the position it was commanded to.

The servomotor accepts pulse width durations of between 1.164 to 2.055 mSecs to control its position over 180 degrees; the motor requires a delay of at least another millisecond before the next pulse

[1] Atmel AVR Microcontrollers (http://www.atmel.com)

[2] 13kg/cm torque servo motor, available from Jaycar.

[3] http://www.interlinkelectronics.com/ FSR Part number 402

can begin. A fixed servo cycle of approximately 333 Hz (3 mSec cycle) is used.

Using the 16-bit interrupt-driven timer port to control the PWM allows the program code to use the entire servo cycle to perform serial transmission with the Java engine; this includes both transmitting the sensor value and receiving an updated motor position (Fig. 10).

Sensor data is transmitted to the desktop Java application at the beginning of a servo cycle, allowing the full cycle to be used for calculations and preparation of a new motor position command, in the form of a pulse width for the next cycle.

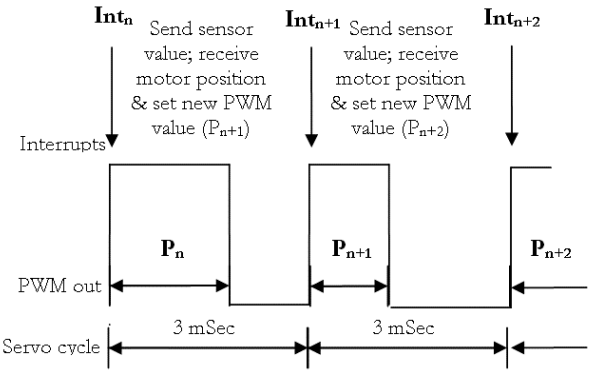


Figure 10. Timing diagram demonstrating integrated interrupt-driven PWM output and motor position calculations.

Mechanical Construction

The carillon allows the player a mechanical advantage over the clapper by employing the principles of torque and rotational force (Fig. 12). In the prototype haptic carillon (Fig. 11), the opposite effort has been made: to give the motor a mechanical advantage over the user.

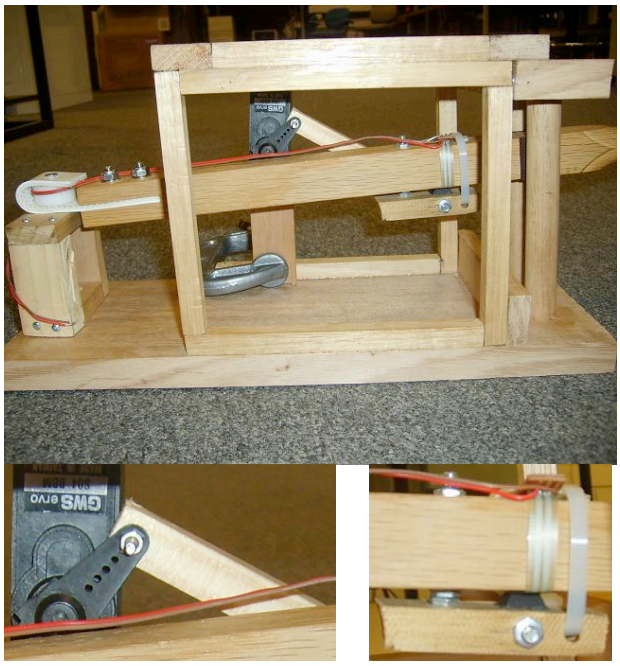


Figure 11. Prototype jig with motor and force sensor attached; b) close-up of the motor arm; c) close up of sensor attachment.

A video of this in action can be viewed at http://www.uow.edu.au/~mh675/baton2.wmv.

Although the baton appears to travel linearly, the mechanism is in fact a rotational one; this alters the amount of force the clapper applies to the bell than if the system were linear. The torque advantage is a function of the distance between the application of force and the pivot point, and the angle at which the force is applied; in the carillon's case the angle is the same for the clapper and the player, allowing the advantage to be expressed as L_P / L_C.

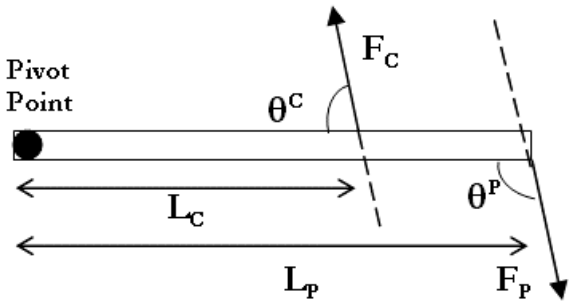


Figure 12. The player has the advantage of applying force over a greater distance to the pivot point than the clapper does; no further advantage is obtained by the angle at which forces are applied, as the angles θ^C and θ^P are complementary.

In the haptic carillon, the motor is coupled to the baton through a stiff wooden component which is free to rotate at both its connection to the motor arm and the baton. This configuration changes the angle at which the user applies force to the baton, and takes advantage of the torque law when angles are involved:

$$Torque = r * F * \sin\theta \qquad (8)$$

where r is the distance from the pivot to the point at which force is applied, and θ is the angle at which force is applied. This wooden link applies a force at an angle which reduces the torque the user can apply to the motor, whilst allowing the motor arm to rotate only a minimal distance to move the baton through its full trajectory.

Conclusion

This paper describes the design and construction of a prototype haptically rendered carillon baton. Using previously developed kinematic models of a grand piano action, our design has recognised the respective differences in performance technique which warrant a uniquely constructed model of a carillon action.

Also described is a method for the implementation of this model in discrete microcontrollers for cost-effectiveness, compactness and speed.

Developed physical models have been tested in Matlab, Java and realtime hardware implementa-

tions, with results in concord with behaviours observed at the National Carillon, Canberra.

Conclusion

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